Marmara Lectures

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Forced LTI differential equations and the convolution integral

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Consider the forced LTI differential equation

$$\frac{d^n}{dt^n}y(t) + a_{n-1}\frac{d^{n-1}}{dt^{n-1}}y(t) + \ldots + a_2\frac{d^2}{dt^2}y(t) + a_1\frac{d}{dt}y(t) + a_0 y(t) = f(t)$$

Our aim is to construct $y(t)$ fot arbitrary $f(t)$. Assume that the impulse response $h(t)$ of this LTI differential equation is known:

$$\frac{d^n}{dt^n}h(t) + a_{n-1}\frac{d^{n-1}}{dt^{n-1}}h(t) + \ldots + a_2\frac{d^2}{dt^2}h(t) + a_1\frac{d}{dt}h(t) + a_0 h(t) = \delta(t)$$

or,

$$\left(\frac{d^n}{dt^n} + a_{n-1}\frac{d^{n-1}}{dt^{n-1}} + \ldots + a_2\frac{d^2}{dt^2} + a_1\frac{d}{dt} + a_0\right)h(t) = \delta(t)$$

Because the differential equation is time invariant, a time shift in its forcing (input) $\delta(t)$ will generate exactly the same time shift in its response (output) $y(t)$. Assuming a time shift of s

$$\left(\frac{d^n}{dt^n} + a_{n-1}\frac{d^{n-1}}{dt^{n-1}} + \ldots + a_2\frac{d^2}{dt^2} + a_1\frac{d}{dt} + a_0\right)h(t-s) = \delta(t-s)$$

Multiply both sides with $f(s)$. As s and t are different, time derivatives will not affect $f(s)$.

$$\left(\frac{d^n}{dt^n} + a_{n-1}\frac{d^{n-1}}{dt^{n-1}} + \ldots + a_2\frac{d^2}{dt^2} + a_1\frac{d}{dt} + a_0\right)h(t-s)f(s) = \delta(t-s)f(s)$$

Let us integrate both sides in s:

$$\int_{s=-\infty}^{\infty}\left[\left(\frac{d^n}{dt^n}+a_{n-1}\frac{d^{n-1}}{dt^{n-1}}+\ldots+a_2\frac{d^2}{dt^2}+a_1\frac{d}{dt}+a_0\right)h(t-s)f(s)\right]ds=\int_{s=-}$$

Again, as s and t are independent, the integration in s can jump over the derivatives in t:

$$\left(\frac{d^n}{dt^n}+a_{n-1}\frac{d^{n-1}}{dt^{n-1}}+\ldots+a_2\frac{d^2}{dt^2}+a_1\frac{d}{dt}+a_0\right)\left[\int_{s=-\infty}^{\infty}h(t-s)f(s)ds\right]=\int_{s=-}^{\infty}$$

Also, remembering the definition of impulse/dirac delta function tells us that

$$\int_{s=-\infty}^{\infty}\delta(t-s)f(s)ds=f(t)$$

Substituing this into the above result gives

$$\left(\frac{d^n}{dt^n}+a_{n-1}\frac{d^{n-1}}{dt^{n-1}}+\ldots+a_2\frac{d^2}{dt^2}+a_1\frac{d}{dt}+a_0\right)\left[\int_{s=-\infty}^{\infty}h(t-s)f(s)ds\right]=f(t)$$

We can immediately deduced that

$$y(t)=\int_{s=-\infty}^{\infty}h(t-s)f(s)ds$$

which is the solution. This integral is known as the **convolution integral**.

which can be factorized as

$$(D-\alpha_1)^{m_1}\ldots(D-\alpha_{k-1})^{m_{k-1}}(D-\alpha_k)^{m_k}y=f(t)$$

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